The Community Psychologist Scheme in Apulia: An Integrated Health Technology Assessment Economic Evaluation

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Rendered in formal British academic English (Oxford house style: plain syntax, measured hedging, minimal use of the first person, British spelling conventions).

Structured Abstract

Background and objective. In Apulia, approximately 780,000–1,160,000 people live with clinically significant psychological distress each year that current primary care does not systematically identify. Regional Law No. 11/2023 established the *Psicologo di Base* ("Community Psychologist") scheme, the first of its kind in Italy. This study evaluates the expected economic and health value of the scheme at full implementation, to inform the intensity and pace of its roll-out.

Methods. An integrated economic evaluation was conducted using a hybrid framework combining four complementary methodological traditions: the EUnetHTA HTA Core Model (nine domains), the Five Case Model, the Consolidated Health Economic Evaluation Reporting Standards (CHEERS 2022), and OECD Development Assistance Committee (OECD-DAC) evaluation criteria. A dual perspective was adopted: the regional health service budget for direct effects, and society as a whole for indirect effects. Three coverage scenarios were compared (620, 775 and 900 Community Psychologists) against an estimated regional steady-state requirement of 800–880 professionals. The time horizon was a static steady-state analysis with a ten-year projection using a Markov model and probabilistic sensitivity analysis. The principal economic anchor is the 2026 OECD methodological revision, explicitly declared to prevail over the preceding anchor (Chisholm et al., 2016) wherever the two diverge.

Results. Under the intermediate coverage scenario (775 professionals), an annual steady-state cost of €40.5 million corresponds to €47.0 million in direct healthcare savings and €27.0 million in indirect benefits — a gross benefit-cost ratio of 6.6:1 — with a cost of approximately €9,800 per healthy life-year gained, well below the Italian cost-effectiveness threshold of €40,000. The direct fiscal account is positive; the health account is the most robust component of the verdict, grounded in evidence of moderate-to-high certainty; the social account is broadly favourable. The three coverage scenarios differ in intensity, not in the direction of the judgement, which is favourable in all three.

Conclusions. The verdict is tripartite and clearly favourable across all three dimensions — fiscal, health, and social — deliberately never aggregated into a single figure. Full implementation of Regional Law No. 11/2023 is recommended under the intermediate coverage scenario, with staggered roll-out across Apulia's six Local Health Authorities. The principal declared limitations concern the lack of statistical significance of direct healthcare savings in the reference international trial, and an editorial reconciliation, still in progress, between the most recent economic anchor and the detailed calculations reported in the analytical sections of the integral volume. Neither limitation alters the direction of the judgement.

Keywords: Community Psychologist; primary care; mental health; economic evaluation; Health Technology Assessment; CHEERS 2022; EUnetHTA; cost-effectiveness; QALY; Apulia.

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1. Introduction

The population of Apulia is both shrinking and ageing: 3,890,661 residents at the end of 2023, a decline of 17,022 on the previous year, with an ageing index that has risen to 200.8 older people for every 100 young people, and a fertility rate of 1.16 children per woman. Among the older population, now approximately 945,000 people, the international epidemiological literature estimates between 95,000 and 142,000 cases of clinically significant depression and between 142,000 and 190,000 cases of anxiety disorders; by 2050, with the population aged over 65 projected to reach 1,200,000, the number of cases is expected to grow by approximately half. These figures are compounded by economic and social conditions that amplify risk: an at-risk-of-poverty-or-social-exclusion rate of 37.7 per cent — the fourth highest in Italy — and youth unemployment of 32 per cent, in a context where the literature documents a bidirectional causal relationship between socioeconomic deprivation and common mental disorders.

Apulian primary care currently addresses this burden without a structured psychological presence at the point of first contact. The region's six Mental Health Departments, comprising forty-seven Mental Health Centres and 1,813 staff, care for 54,946 users, but are predominantly oriented towards severe and persistent conditions rather than the common distress that originates in primary care. In 2023, there were 18,753 emergency department presentations for psychiatric causes — more than fifty per day, a 30 per cent increase on pre-pandemic levels — a signal of demand that is not adequately identified upstream. Applying current international prevalence estimates (20–30 per cent with clinically significant distress in a given year) to the Apulian adult population (approximately 3.9 million people), the study estimates that between 780,000 and 1,160,000 people are affected, against a specialist network sized for only a fraction of this figure.

Regional Law No. 11/2023 established the Community Psychologist scheme, making Apulia the first Italian region to do so. Because the scheme is already established in law, the evaluative question is not whether it should exist, but rather what the economic and social consequences of its steady-state sizing will be: which level of coverage to adopt, and at what implementation pace. This study quantifies the expected value of the scheme to inform the intensity and timing of its implementation.

2. Methods

2.1 Study design and methodological framework

This is a secondary-data synthesis economic evaluation, conducted by a single author, combining four distinct and complementary methodological traditions. The EUnetHTA HTA Core Model provides the overarching grid within which the problem is organised across nine domains: current health problem and use, technical description of the technology, safety, clinical effectiveness, costs and economic evaluation, ethical analysis, organisational aspects, social and patient aspects, and legal aspects. The Five Case Model, developed in the United Kingdom for the appraisal of public investment, requires the joint articulation of the strategic case, the economic case, the commercial case, the financial case, and the management case.

The Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 govern reporting transparency. OECD-DAC evaluation criteria (relevance, coherence, effectiveness, efficiency, impact, sustainability) extend the analysis to the long-term sustainability of the reform.

2.2 Population, intervention, comparator, perspective and outcomes

The reference population comprises approximately 3.89 million Apulian residents, of whom between 780,000 and 1,160,000 present with clinically significant common psychological distress. The intervention under evaluation is the Community Psychologist scheme at full implementation, considered together with the network of directly and indirectly connected professional figures. The sole comparator is the status quo, namely the absence of a structured scheme at steady state. Outcomes considered belong to three distinct categories, never aggregated into a single figure: direct fiscal outcomes (regional health service budget), health outcomes (healthy life-years gained), and indirect social outcomes (productivity, employment, welfare). The perspective is dual: the regional health service budget for direct quantities, and society as a whole for indirect ones. The time horizon is a static steady-state analysis, with a ten-year projection using a Markov model and a discount rate of 3 per cent.

2.3 The organisational model

The economic value of the reform does not originate from a single professional figure, but from a five-tier pyramidal hierarchy, ordered by influence on the scheme and by capacity for economic return, whether direct or indirect. The ordering principle prevents double-counting: each economic effect is attributed to the tier at which it is generated. The first, apical tier is occupied by the Community Psychologist, with a coverage ratio of one professional per 4,400–4,900 residents. The second tier comprises first-contact clinical figures — the general practitioner, forming a "clinical pair" with the Community Psychologist within Community Health Houses, the paediatrician, and the neurologist. The third tier comprises continuity and specialist-filtering figures — the family and community nurse, the consulting psychiatrist, and Mental Health Centres. The fourth tier is occupied by the social worker and territorial social services. The fifth tier comprises the remaining figures and non-health domains connected to the sixteen multidisciplinary domains considered by the study. Artificial intelligence, where deployed, does not constitute an autonomous tier of the hierarchy, but a cross-cutting lever subordinate to human oversight and to Regulation (EU) 2024/1689, with a contribution that is isolated and non-additive relative to the savings generated by professional staffing.

2.4 Economic evaluation techniques

The core of the economic evaluation combines non-alternative, complementary methods, each specialised to a determined quantity: cost-benefit analysis, for the net balance and the benefit-cost ratio; cost-effectiveness and cost-utility analysis, for the incremental cost-effectiveness ratio expressed in quality-adjusted life-years (QALYs); budget impact analysis, for the sustainability of the regional health service budget; social return on investment, for social outcomes; Markov models and microsimulation, for the projection of clinical course; and multi-criteria decision analysis, for the composition of economic, clinical, ethical and organisational dimensions. A cross-cutting non-duplication rule governs the calculation throughout: no effect is attributed to more than one category.

2.5 Economic anchor and methodological versioning

The study initially adopted, as its international anchor for the return on mental health investment, the ranges reported by Chisholm and colleagues (2016): a benefit-cost ratio of 2.3 to 3.0:1 for economic benefits alone, and 3.3 to 5.7:1 including health returns. A subsequent methodological governance review replaced this reference with the 2026 OECD framework and the Unisanté-PLIMeC-P study (registry identifier NCT06377189, University of Lausanne), declaring that these figures prevail wherever they diverge from the body of the integral volume. The rationale for the substitution is methodological: Chisholm and colleagues' estimates derive predominantly from low- and middle-income countries, with cost-effectiveness thresholds not fully applicable to the European context, whereas the OECD framework is calibrated to the European primary care setting. The results reported here adopt the 2026 OECD anchor as authoritative, by explicit decision of the author.

2.6 Uncertainty analysis

The uncertainty protocol projects results over time and verifies their robustness through two complementary instruments. The Markov model and microsimulation represent the course of clinical conditions as a sequence of health states and transitions, over a ten-year horizon. Probabilistic sensitivity analysis subjects the results to repeated simulation over distributions of uncertain parameters, producing a distribution of the incremental cost-effectiveness ratio with a probability of cost-effectiveness reported in 88.9 per cent of simulations. The protocol further declares explicit falsification thresholds: the cost-effectiveness conclusion would change were clinical effectiveness to fall below a declared threshold; the conclusion on social return would change were indirect benefits to fall outside the validated range.

3. Results

3.1 Base case: the intermediate coverage scenario

Under the intermediate coverage scenario (775 Community Psychologists), an annual steady-state cost of €40.5 million corresponds to €47.0 million in direct healthcare savings — through reductions in inappropriate pharmaceutical spending, emergency department presentations, and passive health-related mobility — and €27.0 million in indirect benefits, for a total benefit of €74.0 million and a gross benefit-cost ratio of 6.6:1. The cost per healthy life-year gained is approximately €9,800, well below the Italian cost-effectiveness threshold of €40,000 per QALY. The break-even period is estimated at 2.3 years.

3.2 Scenario analysis

The study evaluated the reform under three coverage scenarios — 620, 775 and 900 Community Psychologists — against an estimated regional steady-state requirement of 800–880 professionals. Under the 2026 OECD anchor, direct healthcare savings range from €23.5 million in the minimum scenario to €47.0 million in the intermediate scenario and €79.5 million in the full scenario, with the cost per QALY falling from approximately €11,500 to €9,800 to €7,200 respectively. The three scenarios differ in coverage intensity, not in the direction of the judgement, which is favourable in all three cases. The intermediate scenario is recommended as the point of equilibrium between recruitment feasibility and coverage of the prudential requirement, with staggered deployment across Apulia's six Local Health Authorities and forty-five districts.

3.3 Robustness and uncertainty

Probabilistic sensitivity analysis reports a probability of cost-effectiveness of 88.9 per cent, consistent across all three coverage scenarios. The representative patient in the Markov model presents an expected saving of €1,304 and a gain of 0.236 quality-adjusted life-years.

3.4 Equity

The distributional analysis considers three convergent dimensions: territorial equity across regional areas, equity across populations and social groups, and equity of access related to the potential use of digital tools and artificial intelligence in support of the care pathway.

4. Discussion

4.1 Principal findings

The study produces a tripartite verdict that is clearly favourable, deliberately not aggregated into a single figure so as not to conflate quantities of a different nature. The direct fiscal account — the regional health service budget — is positive. The health account — the ratio of incremental cost to health gain — is the most robust component of the verdict, grounded in clinical evidence of moderate-to-high certainty. The social account is broadly favourable. The independent convergence of the three judgements, each robust on its own terms, is arguably more robust than a single aggregated figure would be, applying the same non-duplication rule that governs the methodological framework as a whole.

4.2 Comparison with existing evidence

The collaborative care model that the general practitioner–Community Psychologist clinical pair translates into the Apulian context is supported by more than ninety international randomised controlled trials, with documented reductions in depressive and anxiety symptoms, emergency department presentations, and psychiatric admissions in comparable implementations across Europe, North America and Australia.

4.3 Strengths

The study combines four complementary methodological frameworks rather than relying on a single one, applies an explicit non-duplication rule to economic effects, declares explicit falsification thresholds rather than relying solely on a point expected value, and is constructed according to a replicable template, already extended to nineteen further Italian regional contexts and to a dedicated section covering the European Union of twenty-seven member states.

4.4 Limitations

The study explicitly declares several limitations, consistent with its own falsifiability constraint. Direct healthcare savings, while present in the reference literature on collaborative care (Unützer et al., 2008), did not reach statistical significance when considered in isolation in the reference international trial; the positive fiscal balance should therefore be read as a prudential estimate, not as a guarantee free of uncertainty. The section of the study devoted to clinical effectiveness and impact mechanisms is, at the time of writing, developed only partially relative to the other sections of the integral volume. Staffing requirements depend on a coverage ratio explicitly declared as a prudential range rather than a point estimate; the feasibility of recruiting 620–900 qualified professionals remains the principal implementation risk. The revision to the 2026 OECD anchor has not yet been editorially reconciled, in full detail, with the granular calculations reported in the analytical sections of the integral volume, which remain partly expressed under the preceding anchor; this reconciliation work, still in progress, does not alter the direction of the judgement, which is favourable under both bases of calculation. The clinical safety domain, among the nine domains of the HTA Core Model, has not been developed as an autonomous domain for the psychological intervention itself, being addressed primarily in relation to the artificial intelligence component. Finally, as a secondary-data synthesis conducted by a single author, the study does not include a structured patient and public involvement process within its own methodological design.

5. Conclusions

The verdict is tripartite and clearly favourable across all three dimensions considered — fiscal, health and social — deliberately kept distinct rather than aggregated into a single figure. Full implementation of Regional Law No. 11/2023 is recommended under the intermediate coverage scenario (775 Community Psychologists), with staggered roll-out across Apulia's six Local Health Authorities, recruitment via framework agreement, and continuous monitoring through an indicator system structured according to the Donabedian model, supported by counterfactual impact evaluation.

Declarations

Funding. This study received no external funding, public or private; it is an independent, self-funded piece of work.

Conflicts of interest. The author is not registered with the Italian professional register of psychologists (*Ordine degli Psicologi*) and declares no relevant conflicts of interest in relation to this study.

Ethical approval. Not applicable: this study is a synthesis analysis of secondary, aggregate data (regional accounts, scientific literature, international benchmarks), without the collection of primary data on individual subjects.

Patient and public involvement. No structured patient or public consultation was undertaken in the methodological design of the study, consistent with its nature as a secondary-data synthesis.

Data availability. The full analysis, with complete technical documentation, mapping to the HTA Core Model domains, and the CHEERS 2022 compliance checklist, is available in the integral body of the study (Volume I — Apulia) and in the derivative summary products at various levels of detail.

Author contribution. The author was solely responsible for the study design, the collection of secondary sources, the analysis, and the drafting of the manuscript.

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For the complete methodological framework and detailed mapping to the nine HTA Core Model domains, see Level 4 of the information pyramid (`_livelli-piramide/livello-4-sintesi-tecnica.md`); for the full analysis, Volume I (`tomo-1-puglia/opera-integrale-puglia.docx`).